

CS5707 Artificial Intelligence
Image Processing

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Overview

- **Image fundamentals**
- **Face recognition**
- **Image segmentation**
- **Image recognition**

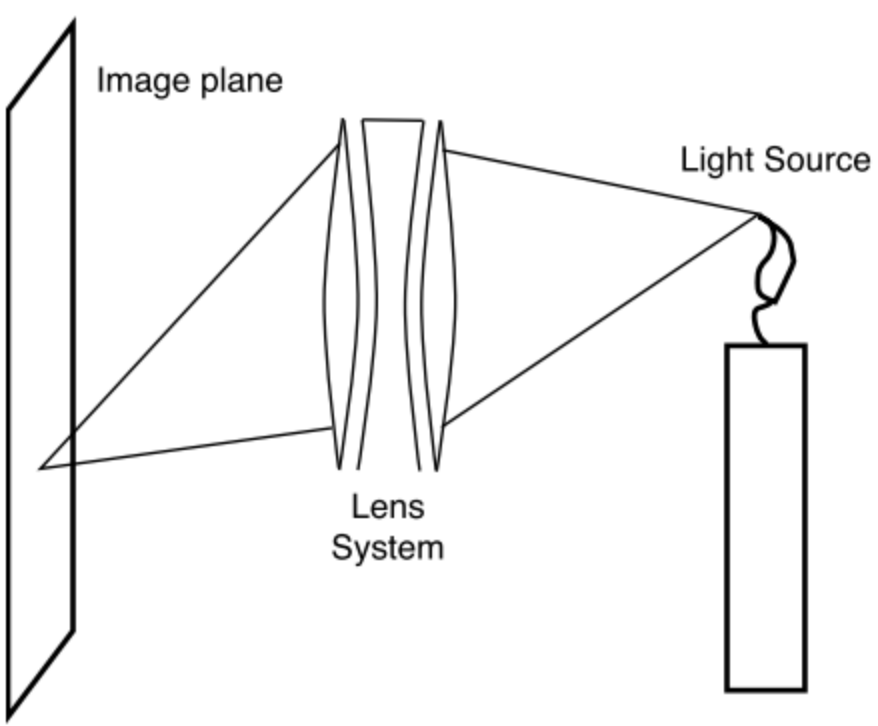
Reading

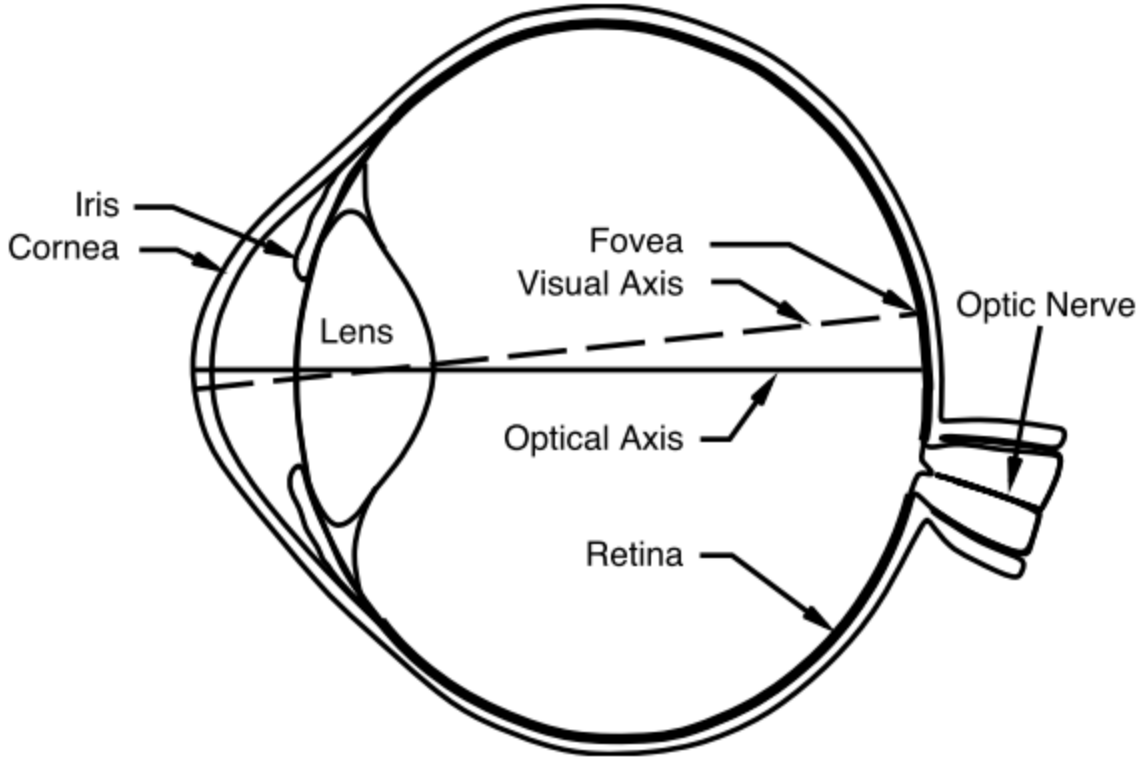
1. Russell, S. and Norvig, P. (2021). §27, Computer vision.
2. Viola & Jones 2001 - Rapid object detection using a boosted cascade of simple features.
3. Long 2015 - Fully Convolutional Networks for Semantic Segmentation.
4. Krizhevsky 2012 - ImageNet Classification with Deep Convolutional Neural Networks.
5. Redmon 2016 - You Only Look Once_ Unified, Real-Time Object Detection.

Image Fundamentals

- **Image formation**
- **Image formats**
  - **binary images**
  - **gray-level images**
  - **colour images**

Image Formation





Lens system of a camera vs human eye

Colour Depth

- **1-bit images**
- **8-bit gray-level images**
- **24-bit colour images**

1-bit Images

- **Each pixel is stored as a single bit (0 or 1), so also referred to as binary image.**
- **Such an image is also called a 1-bit monochrome image since it contains no colour.**



8-bit Gray-Level Images

- **Each pixel has a gray-value between 0 and 255. Each pixel is represented by a single byte; e.g., a dark pixel might have a value of 10, and a bright one might be 230.**



24-bit Colour Images

- **In a colour 24-bit image, each pixel is represented by three bytes, usually representing RGB.**



24-bit Colour Images



(a)



(b)



(c)



(d)

(a): Example of 24-bit colour image “forestfire.bmp”. (b, c, d): R, G, and B colour channels for this image

Face Recognition

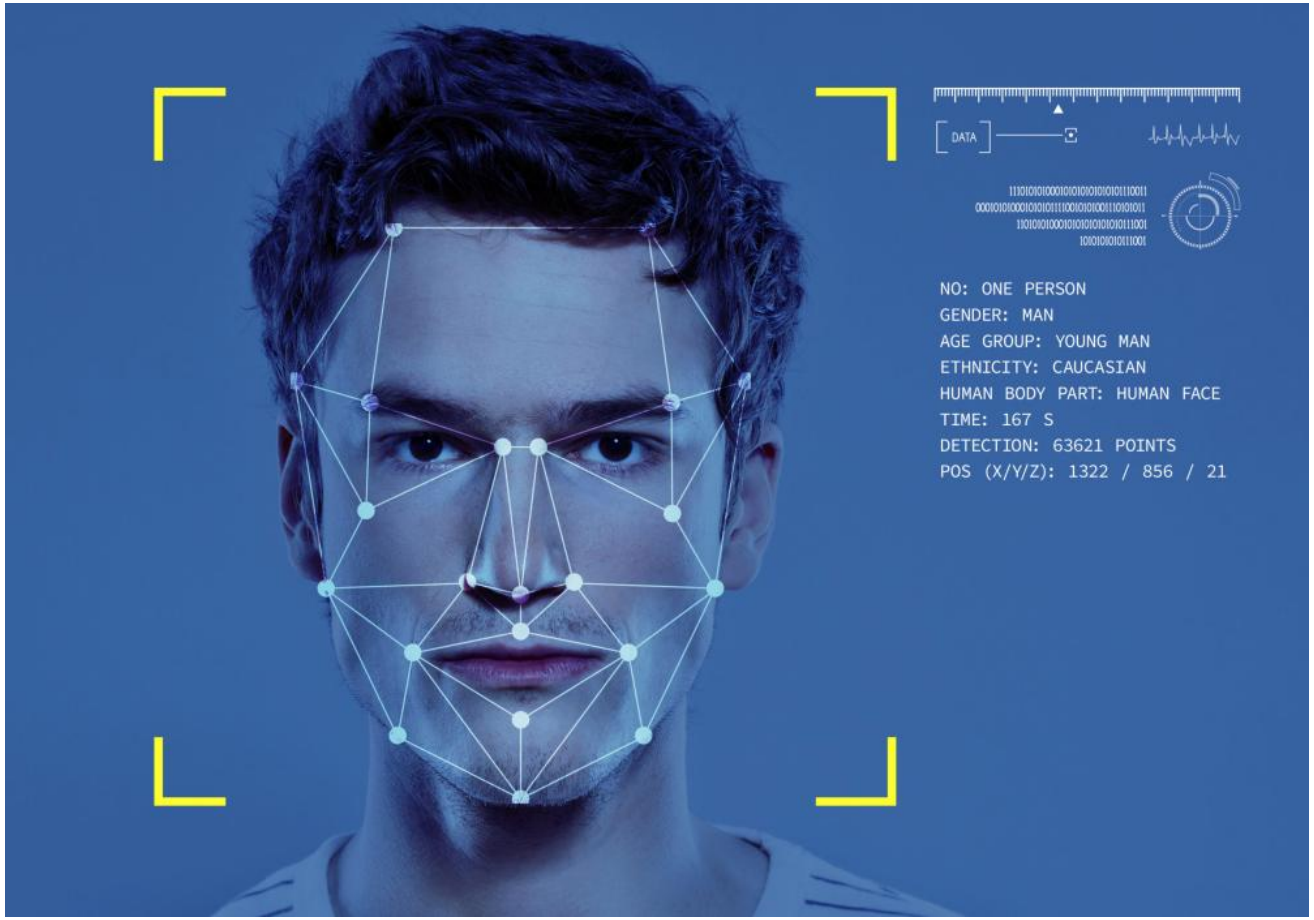
Applications

- **Biometrics: identify verification**
- **Border control**
- **Video surveillance**
- **Apple's Face ID**
- **Facebook's DeepFace:**
  Accuracy of 97.35%
  Human-level performance

Different Problems

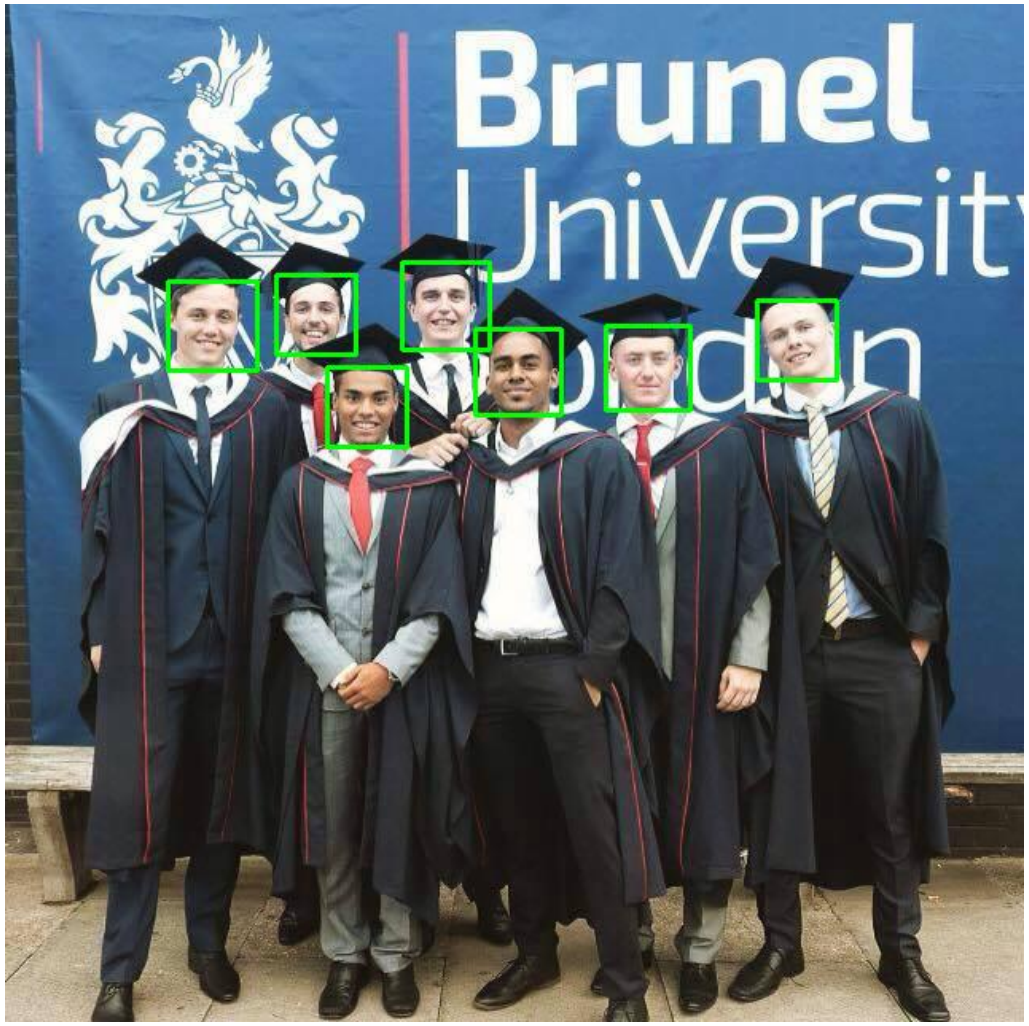
- **Face detection**
- **Verification: 1 vs 1**
- **Recognition: 1 vs many**
- **Expression recognition**
- **Face editing**

Methods



- Find the faces in image
- Locating face landmarks
- Encoding the face
- Train/apply the model

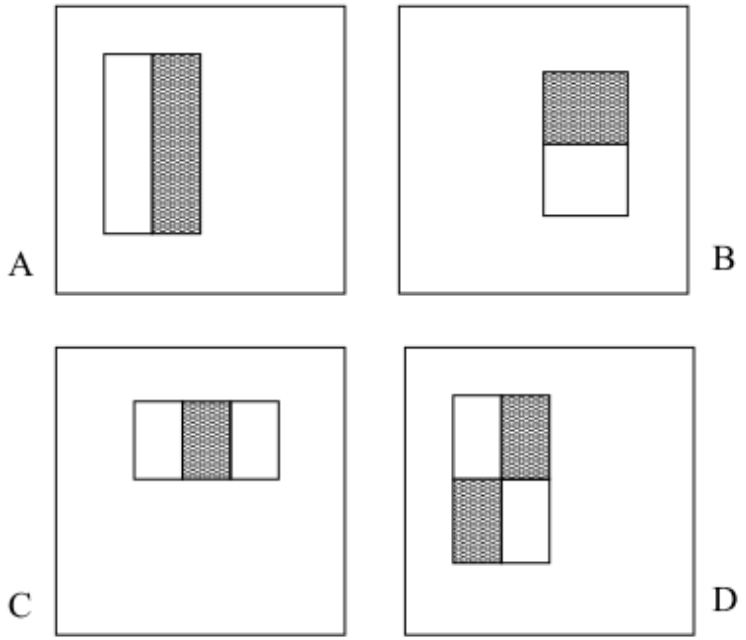
Haar Cascade Face Detector



- First developed in 2001
- By Viola & Jones
- Still one of the fastest face detectors

Detection results using the Haar Cascade face detector.

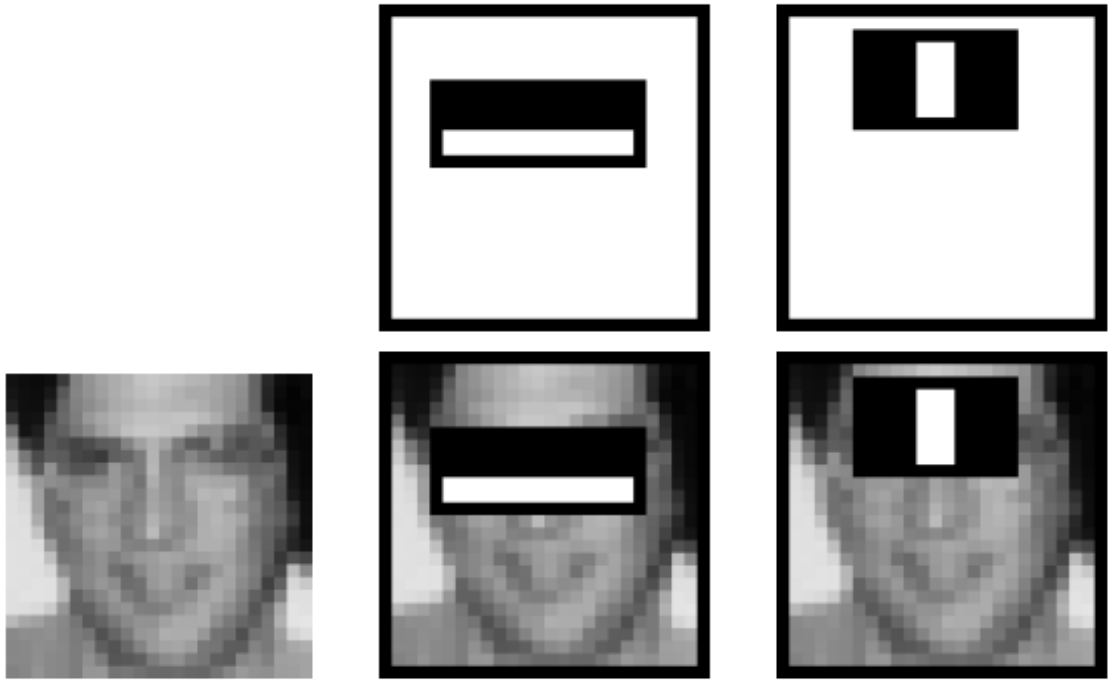
Haar Features



A: 2-rectangle feature
B: 2-rectangle feature
C: 3-rectangle feature
D: 4-rectangle feature (diagonal)

Viola and Jones. Rapid object detection using a boosted cascade of simple features. ICCV 2001.

Haar Features for Face Detection

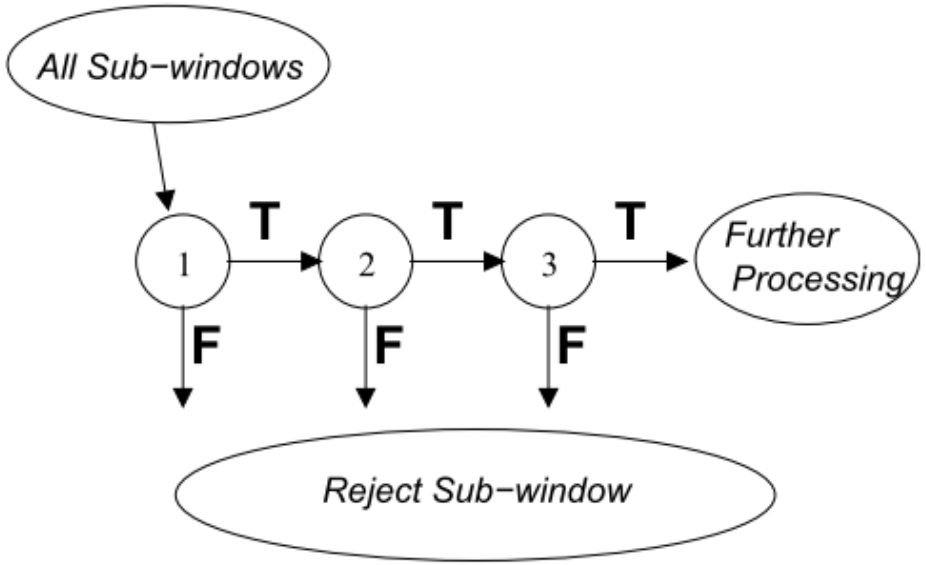


How Haar features work:

A: eye region darker than cheek

B: two eyes darker than nose bridge

Detection Cascade



Cascade: a series of detectors.

- Input: a sub-windows from the scanning
- The first detectors reject the candidates quickly without further processing.
- Only the most likely candidates pass through.

Image Segmentation

Image Segmentation

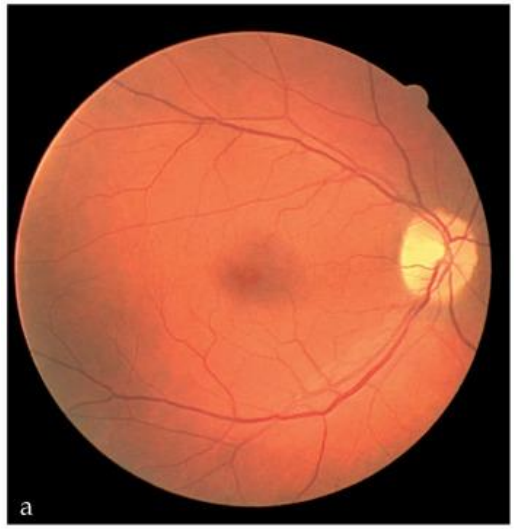
- **Image segmentation:**
  to partition an image into *meaningful* regions
- **The segmentation is based on measurements taken from the image and might be *greylevel*, *colour*, *texture*, *depth* or *motion***

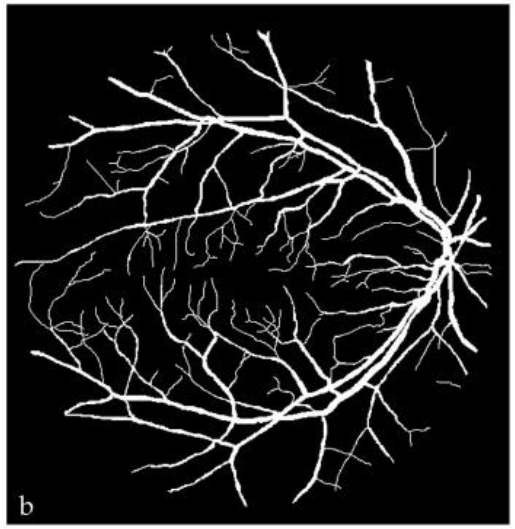
Automated Driving



Segmentation of road surface, vehicles, pedestrians, buildings, trees, traffic lights, signs…

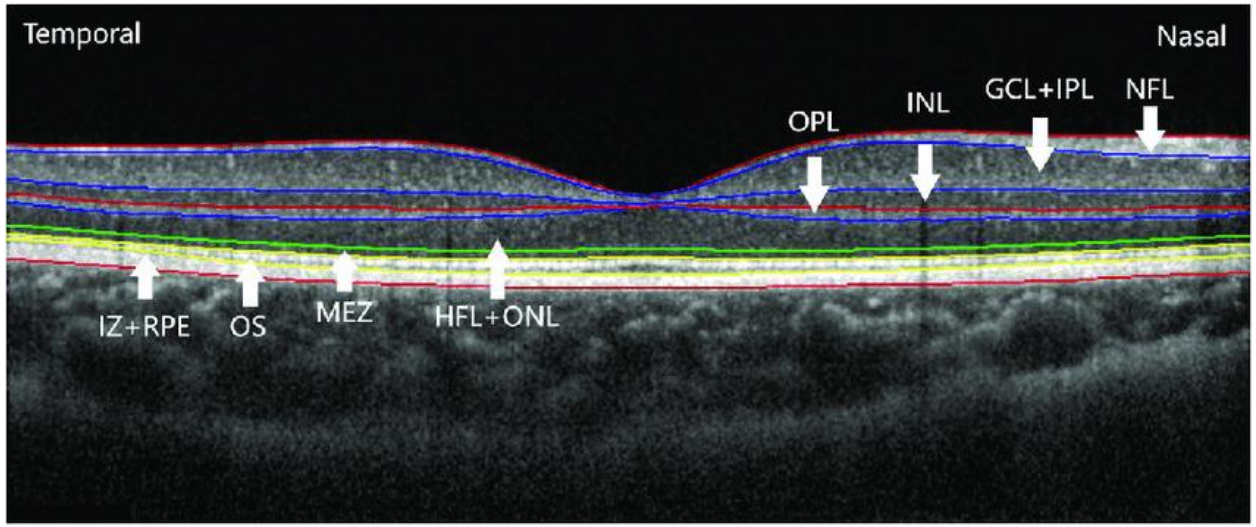
Medical Imaging





Fundus image of retina:

Segmentation of blood vessels and optic disc of the eye



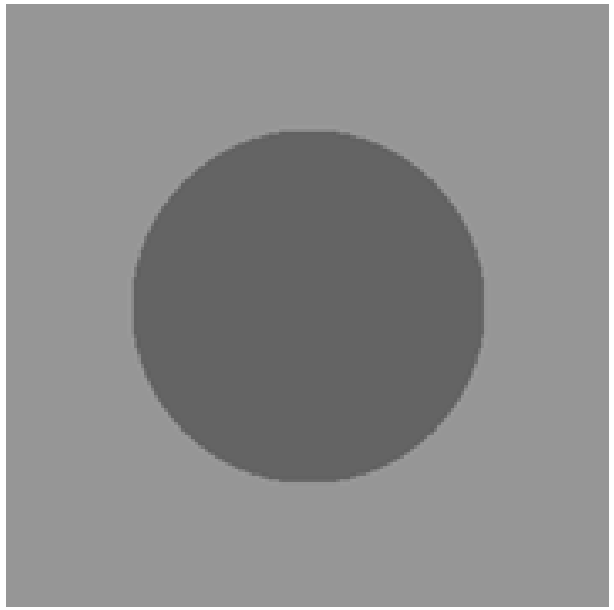
Optical Coherence Tomography (OCT) image:

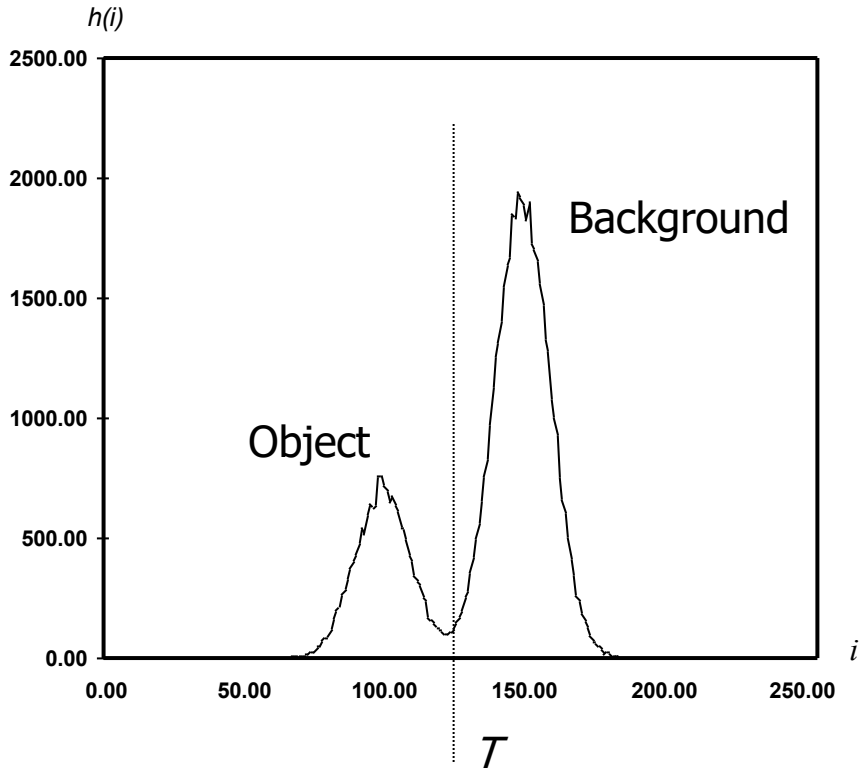
Segmentation of retinal layers

Low-Level Segmentation Methods

- **Grey-levels**
- **Colours**
- **Textures**
- **Region growing**
- **…**

Segmentation by Colours/Greylevels





A dark **object** appearing in a bright **background**

Segmentation by Texture

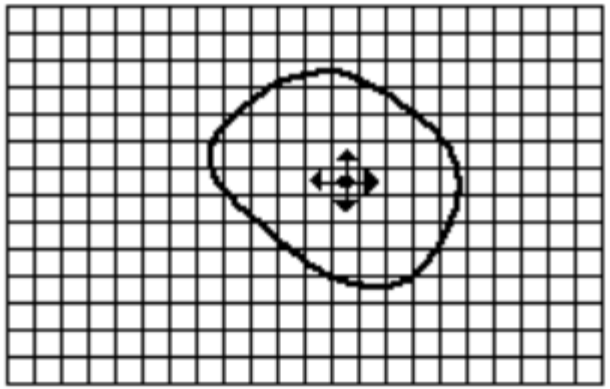


Input image

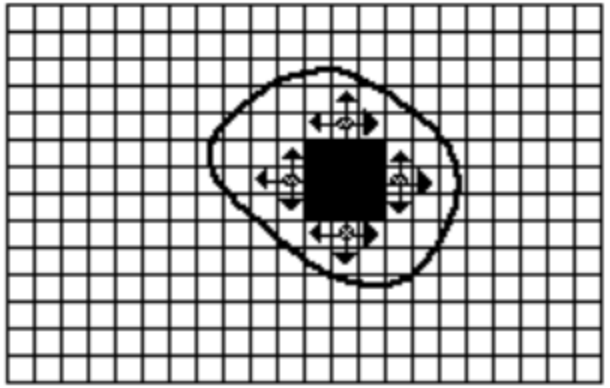


Segmentation result

Segmentation by Region Growing



Initial seed

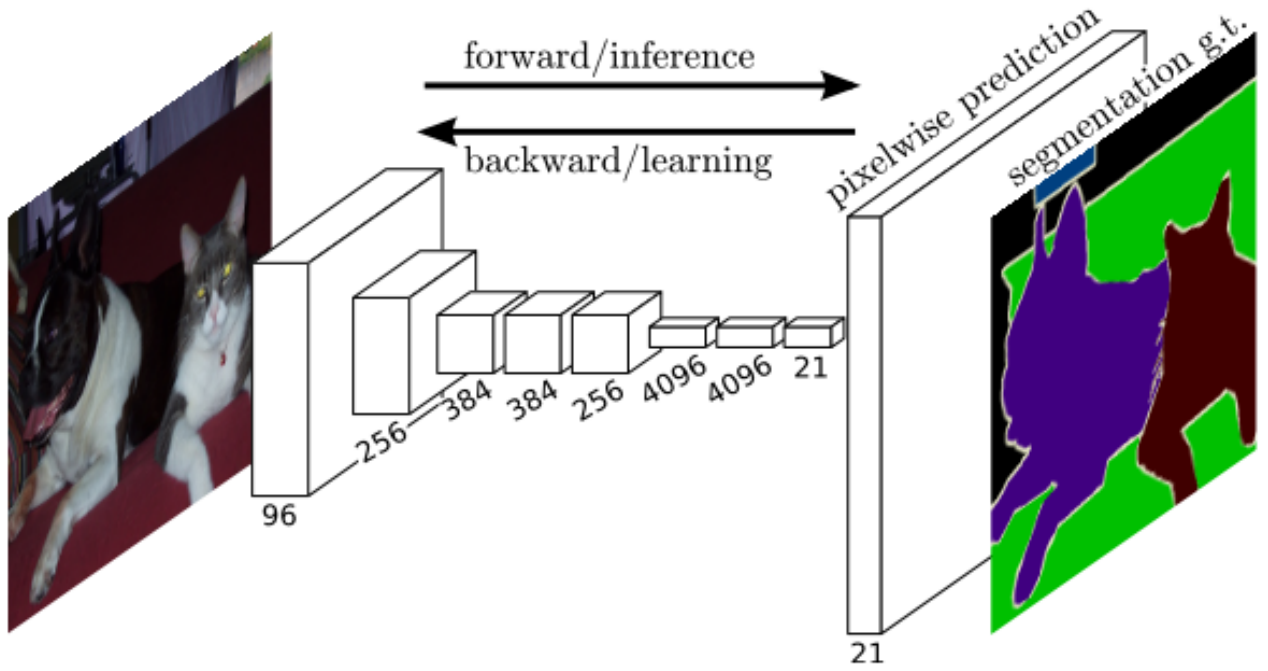


Growing until reaching boundary

High-Level Segmentation Methods

- Clustering
- Compression
- Markov Random Fields (MRFs)
- Graph cut
- Level set
- Neural networks
  - R-CNN
  - Fast R-CNN
  - Faster R-CNN
  - YOLO

Neural Networks for Segmentation



Long et al. 2015. fully convolutional networks for semantic segmentation.

Semantic vs Instance Segmentation



Semantic Segmentation



Instance Segmentation

- **Semantic Segmentation**: only concerns categories - multiple instances of a category as one entity.
- **Instance Segmentation**: distinguish different instances of a same category.

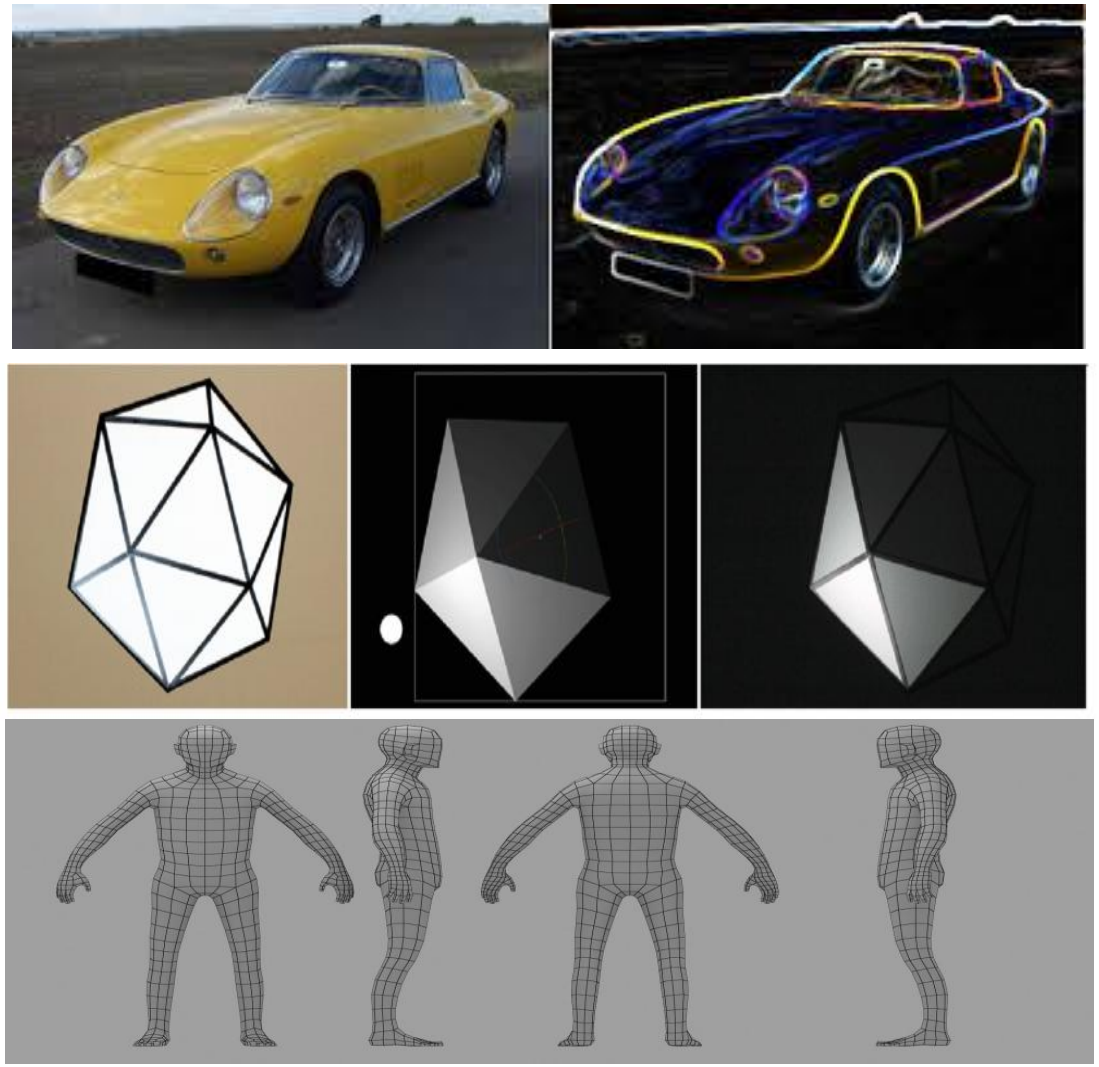
Results of Instance Segmentation



You will try in the lab!

Object Recognition

The Development…

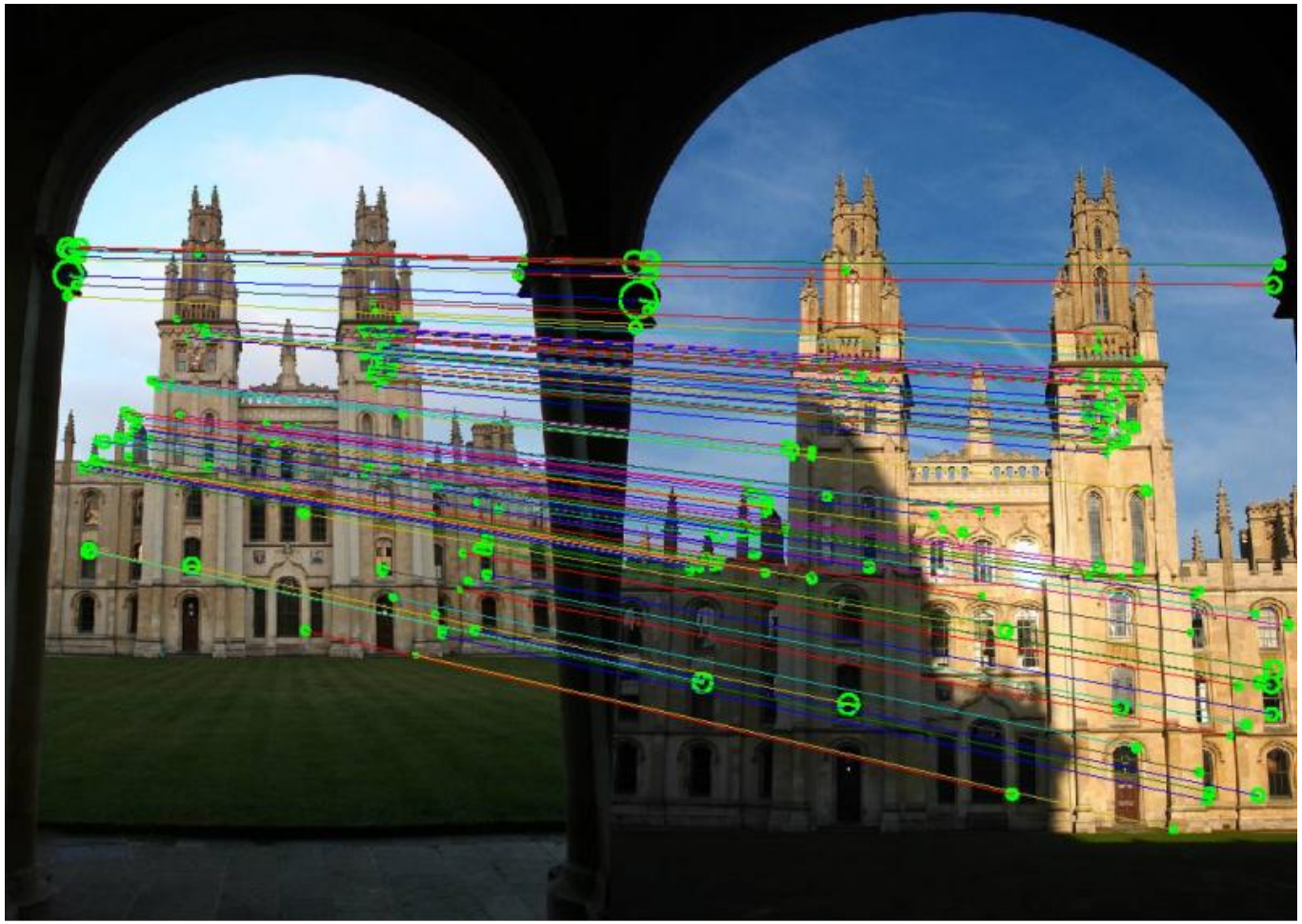


Early stage: edges, boundaries, shape, sketches etc.

2 ½ D: depth, discontinuities, but only projection into the 2D image plane.

Full 3D models.

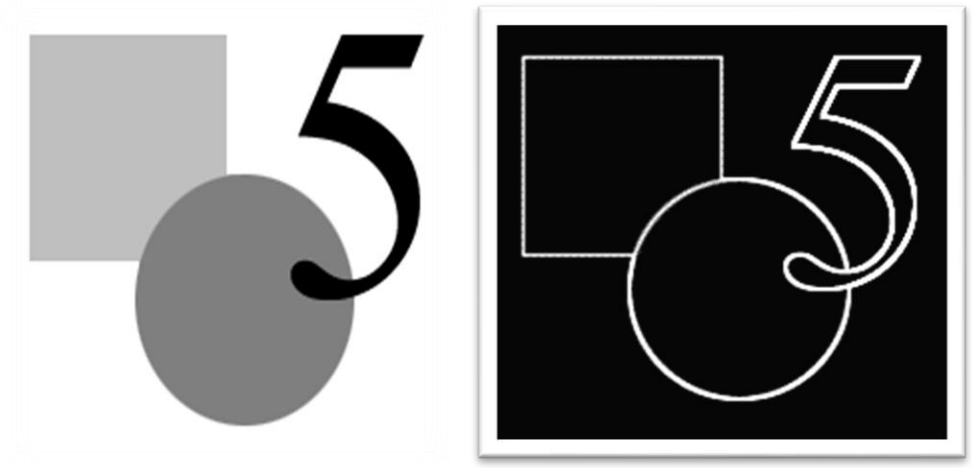
Feature-Based Recognition (1)

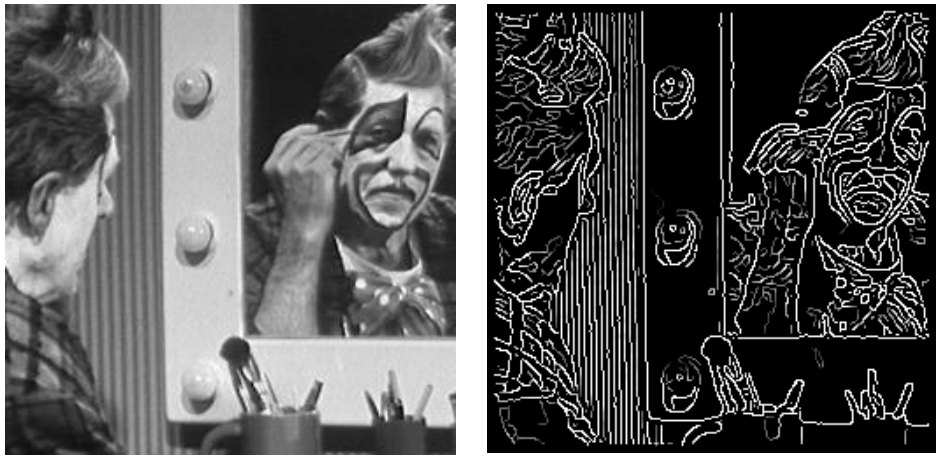


Stage 1: Features extraction

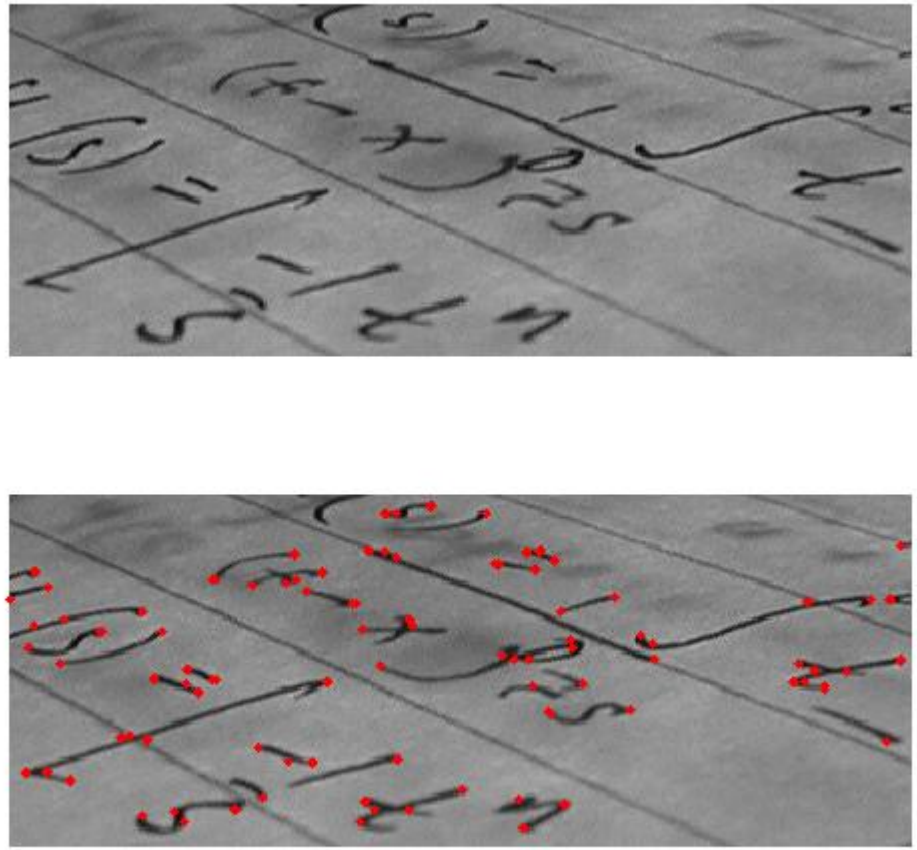
- Local
- Representative
- Robust
- Invariant
- …

Features: Edges



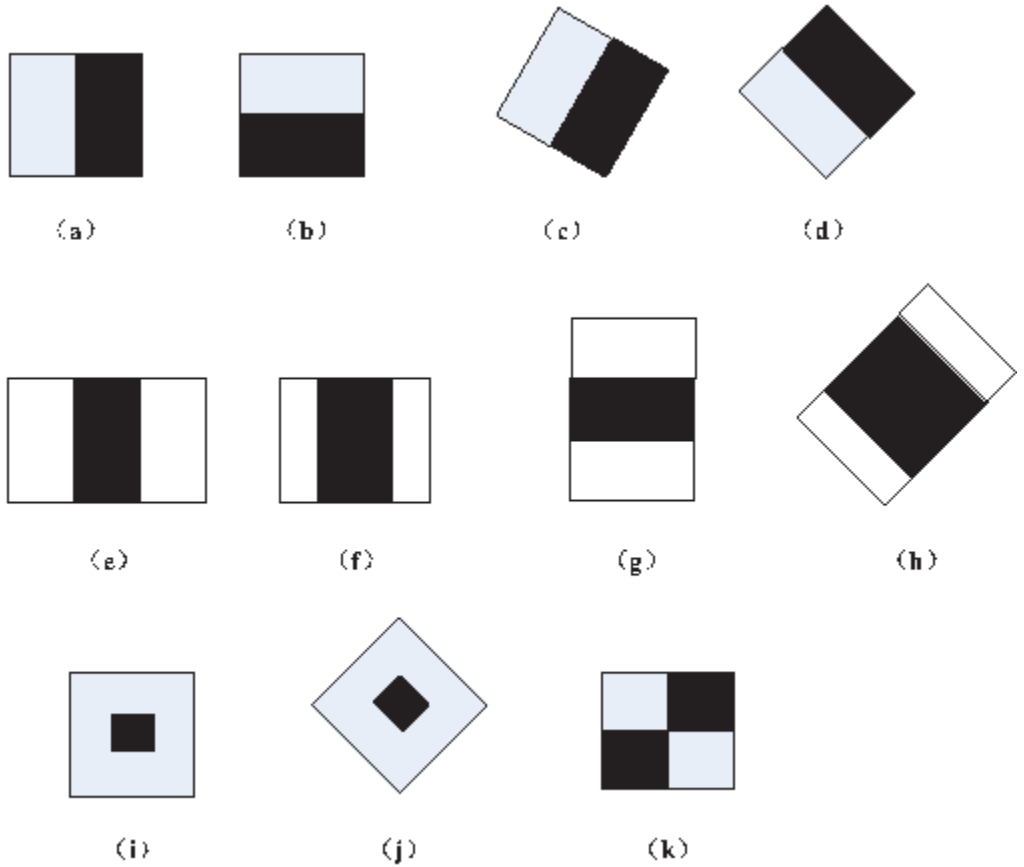


Features: Corners

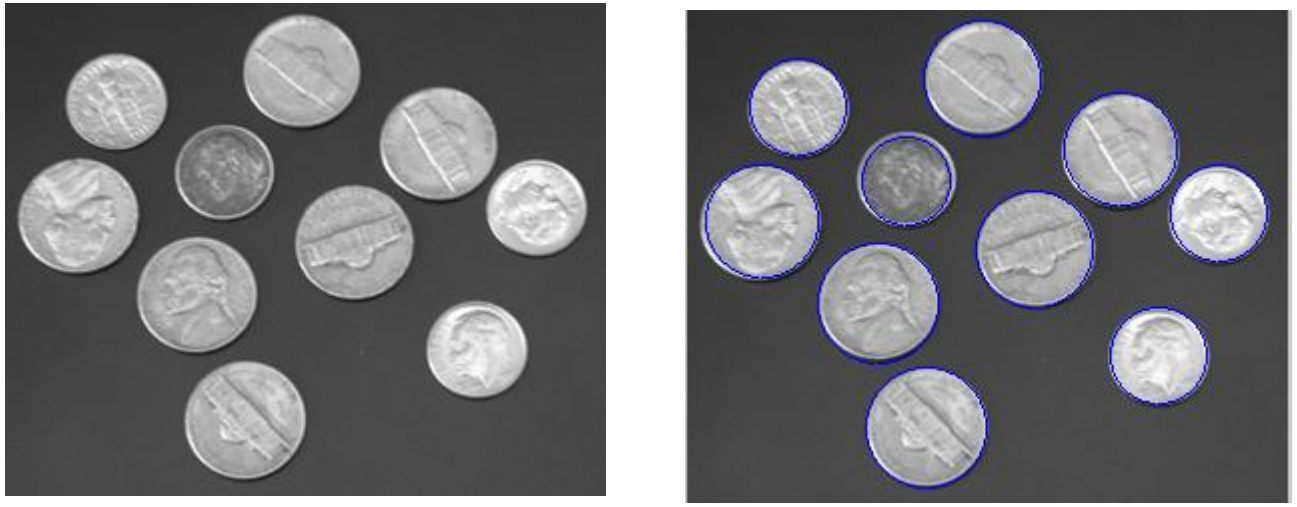


Corners detected by Harris detector.

Features: Haar Features

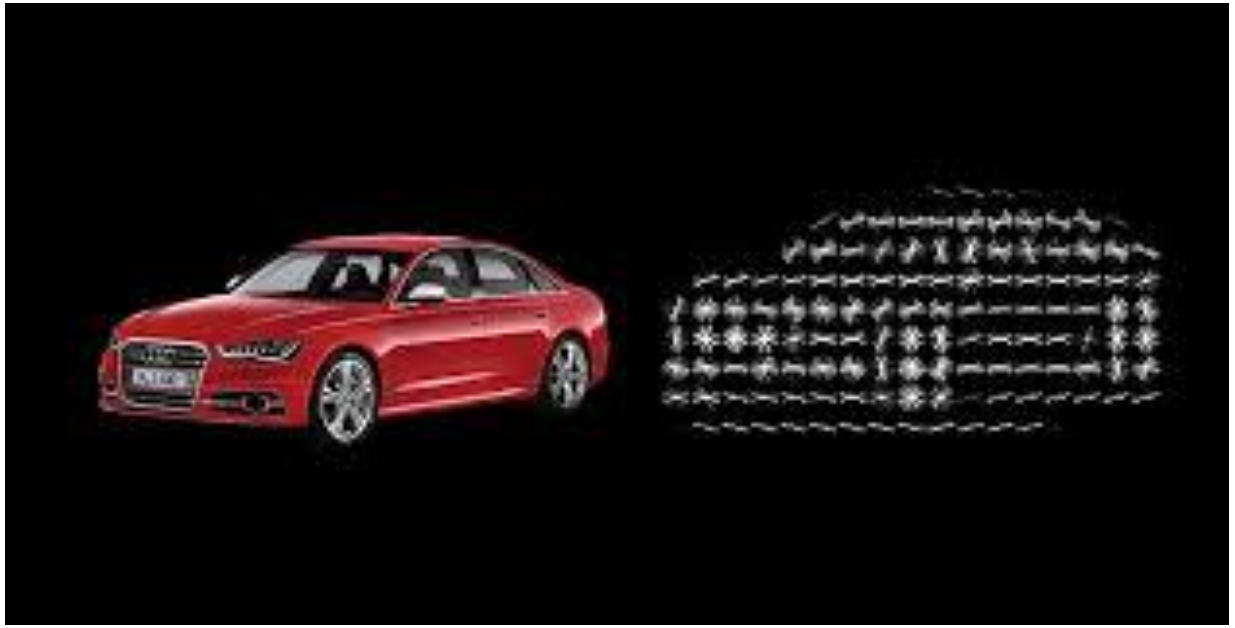


Features: Shapes



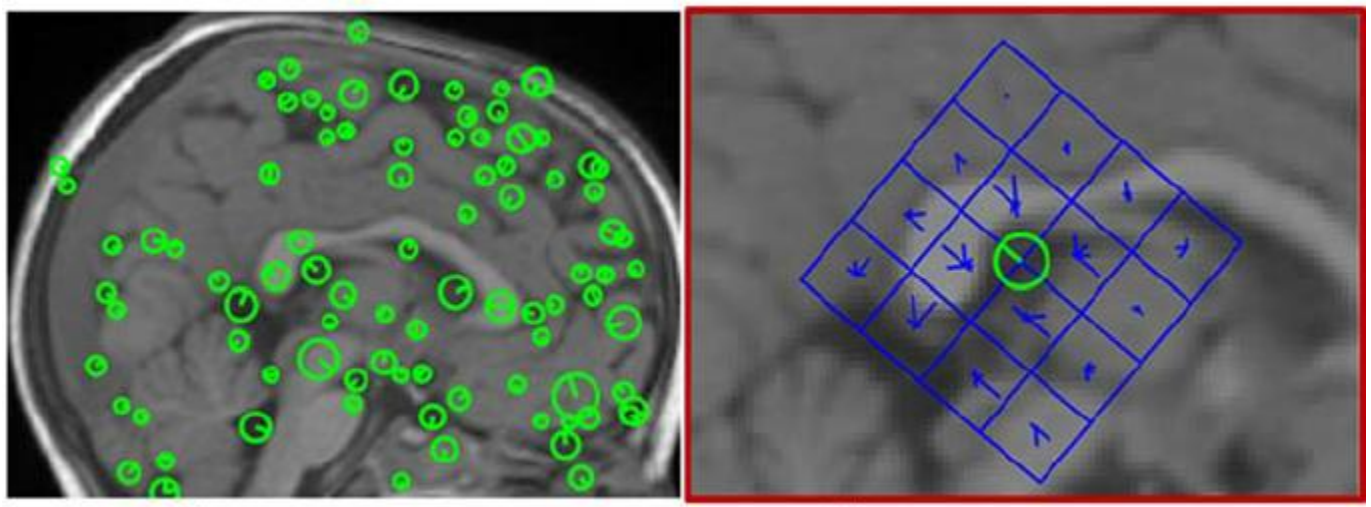
Circles detected by Hough transform.

Features: HOG



Histogram of gradients (HOG)

Features: SIFT



a) SIFT features b) Appearance matrix

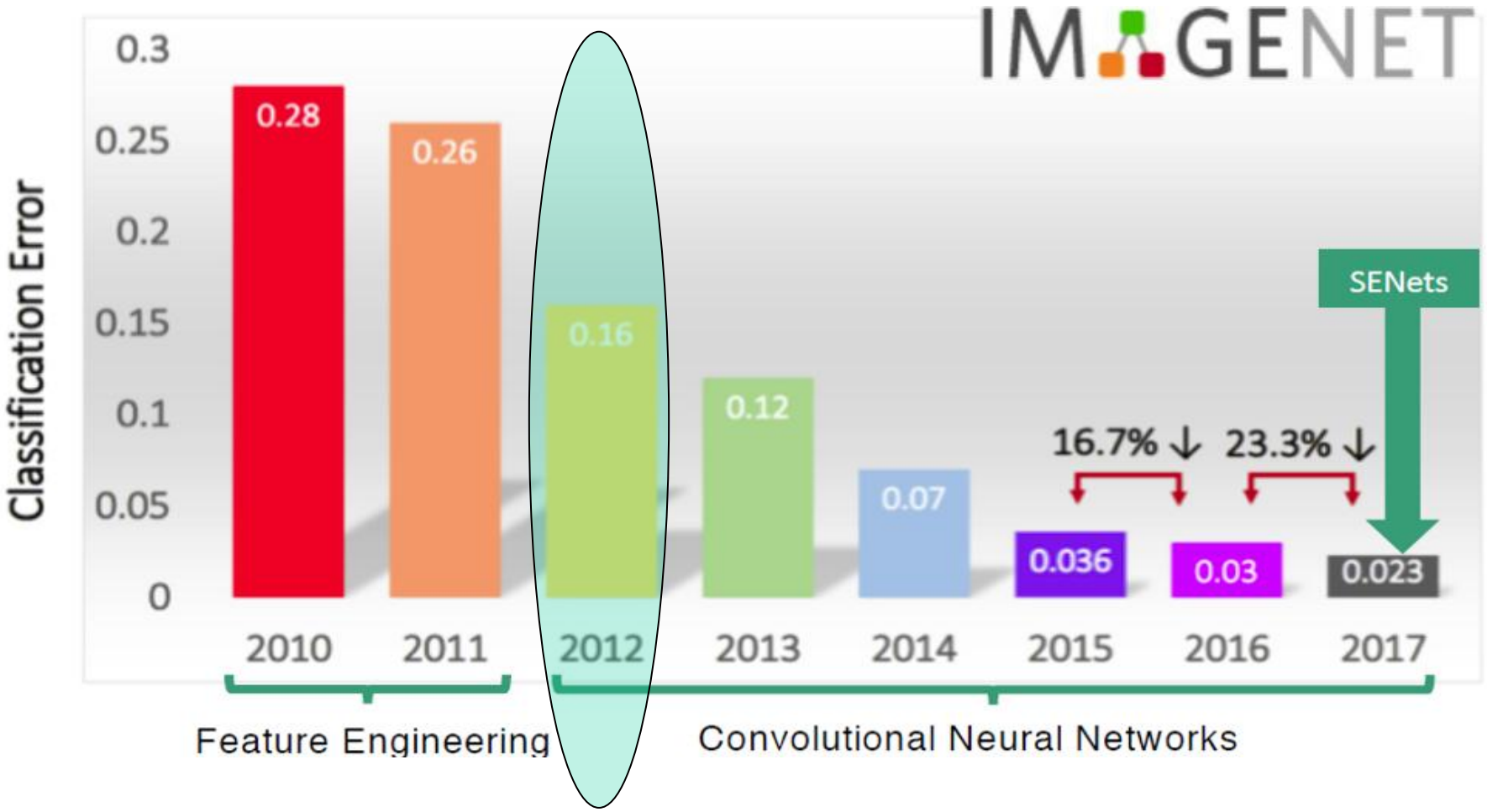
Scale-invariant feature transform (SIFT)

Feature-Based Recognition (2)

Stage 2: Classification

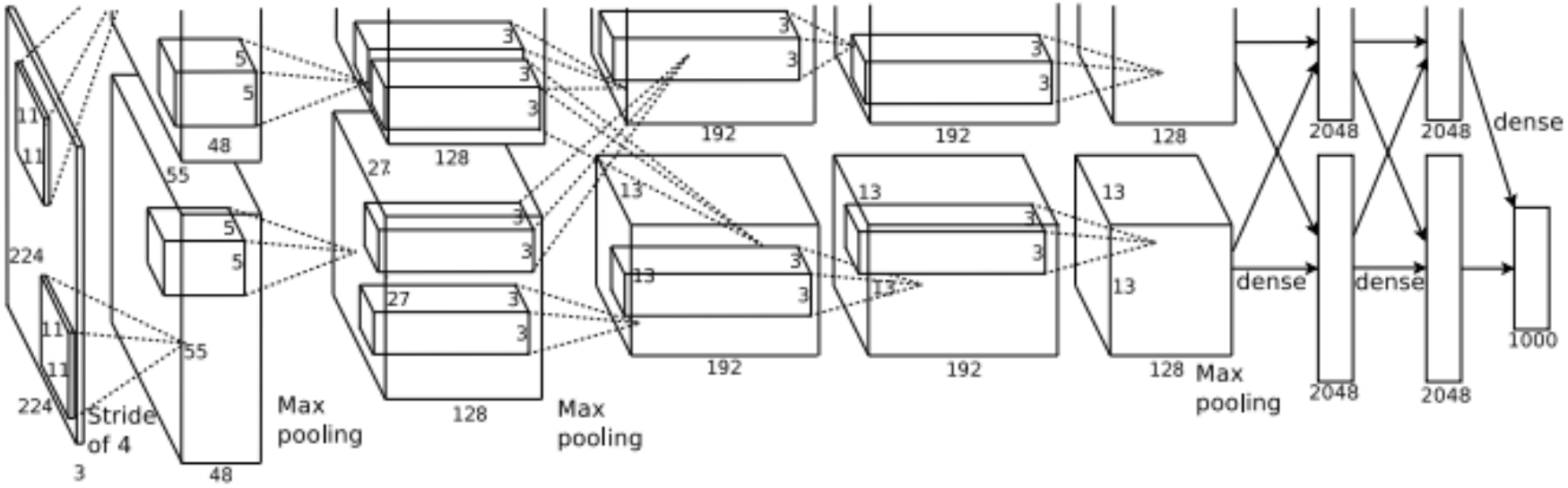
- Bayesian methods
- Genetic algorithms
- Decision trees/forests
- Nearest neighbours
- Support vector machines
- Regression analysis
- Neural networks

ILSVRC



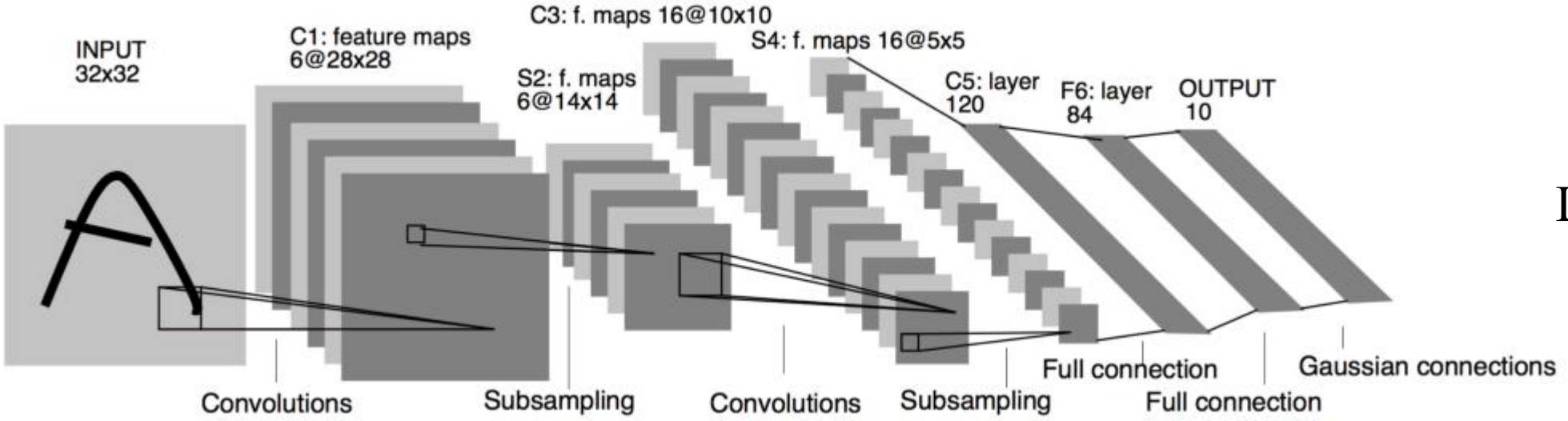
IMAGENET Large Scale Visual Recognition Challenge (ILSVRC)

AlexNet 2012

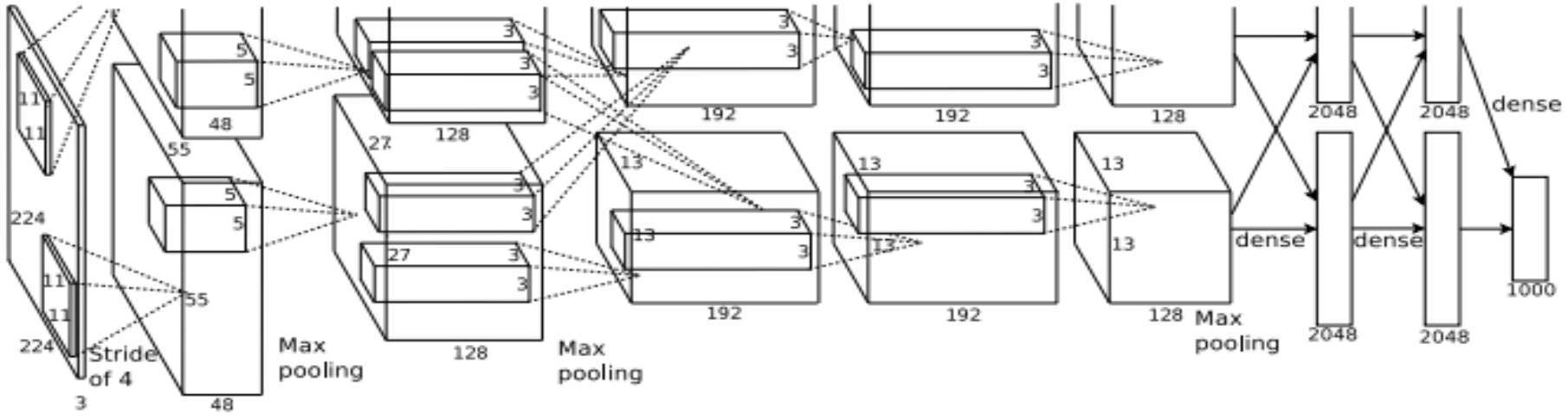


- Winner of ILSVRC 2012
- Alex Krizhevsky
- Error rate of 15.3%, more than 10.8% lower than the runner-up
- One of the most influential work in computer vision
- New era of AI!

LeCun 1998 vs AlexNet 2012



LeCun 1998



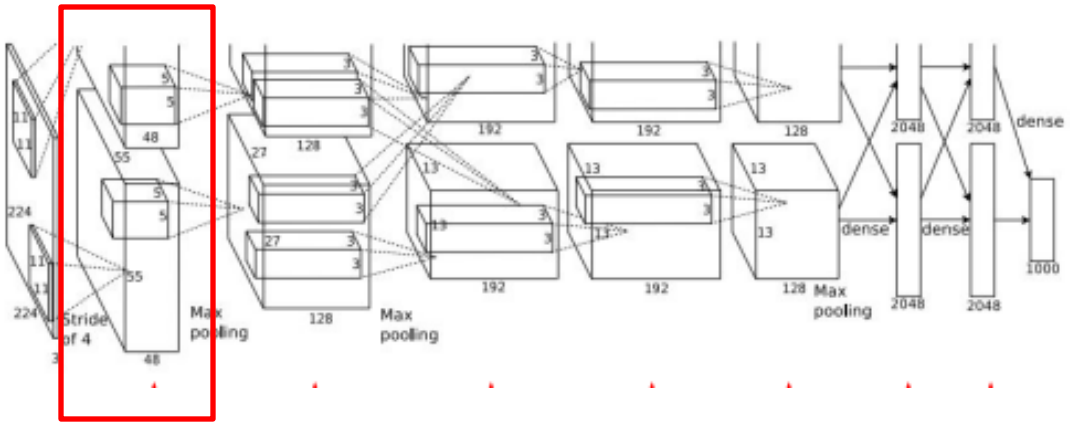
AlexNet 2012

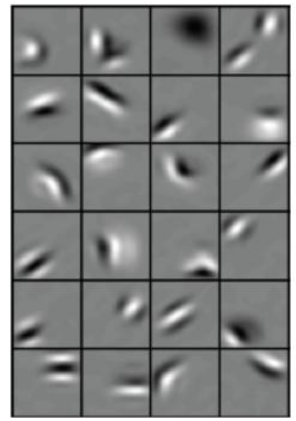
Modern Deep-Learning Recognition

- **A single end-to-end neural network model**
- **No need for feature extraction:**
  first few layers equivalent to feature extraction
- **No need for a classification model:**
  last few layers can approximate any classification functions

Inside the neural networks

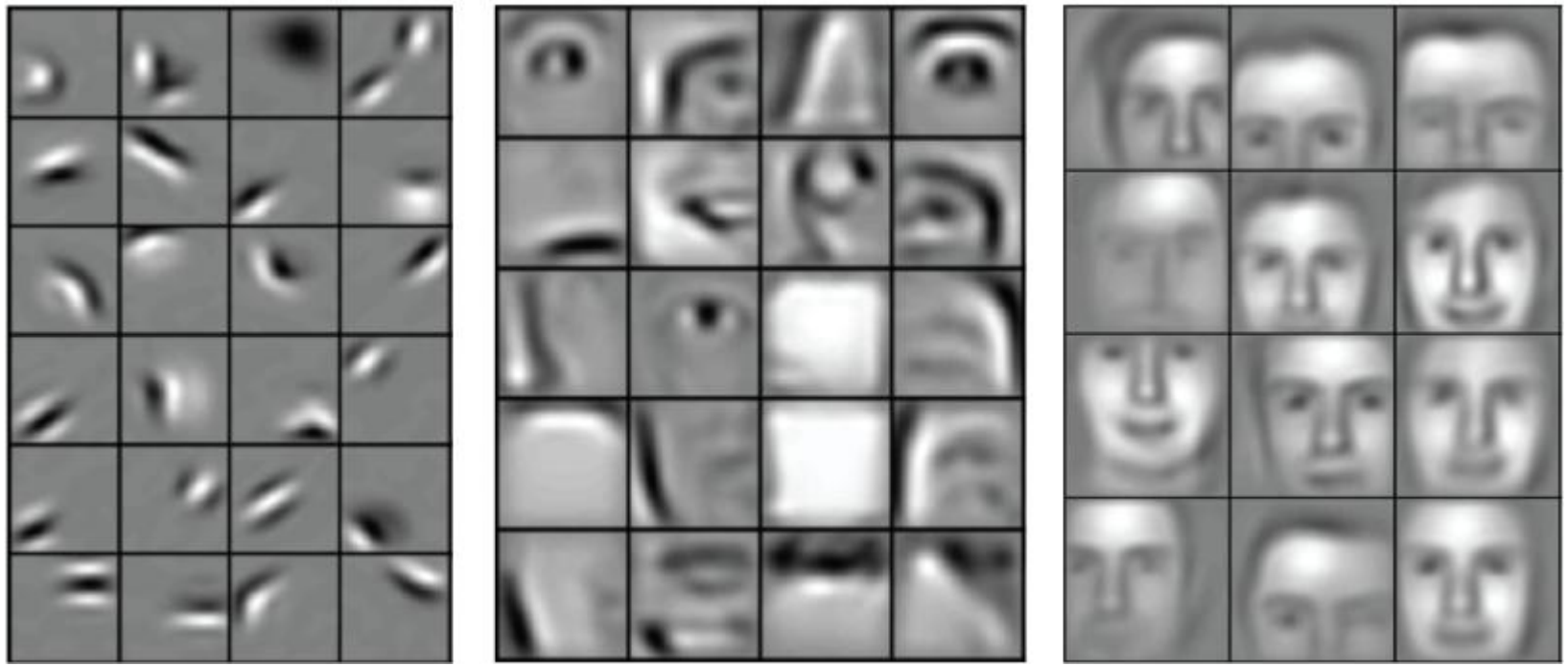
- Visualising the convolution masks in a deep network
- First layer





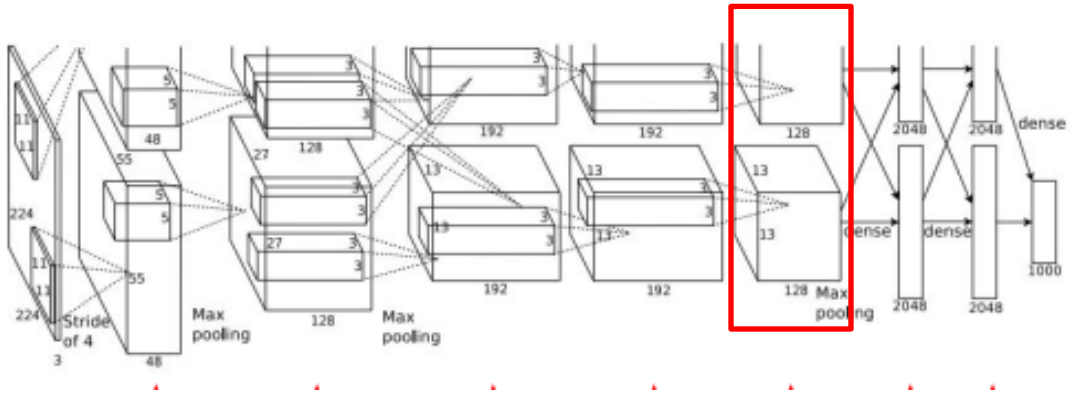
Inside the neural networks

- **From left to right, features of different levels of complexity (Lee 2012)**



Inside the neural networks

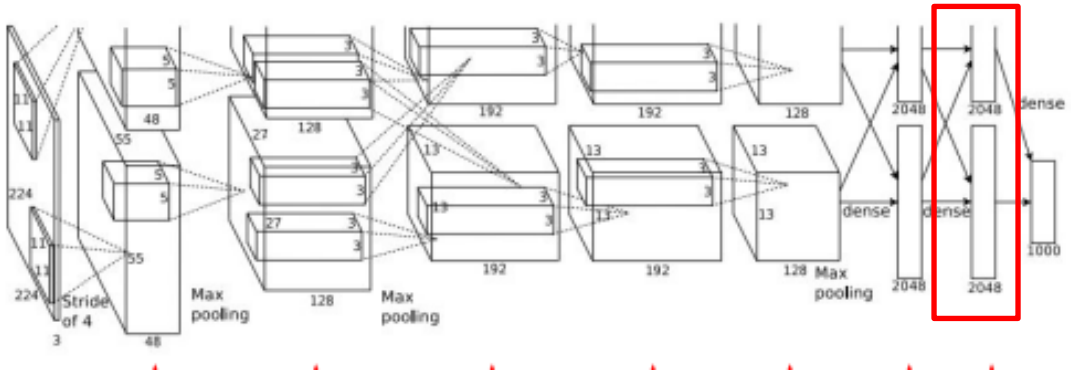
- Highly activating patches
- Pick a layer and a channel
- Run many images through the net
- Visualise the image patches with highest activations

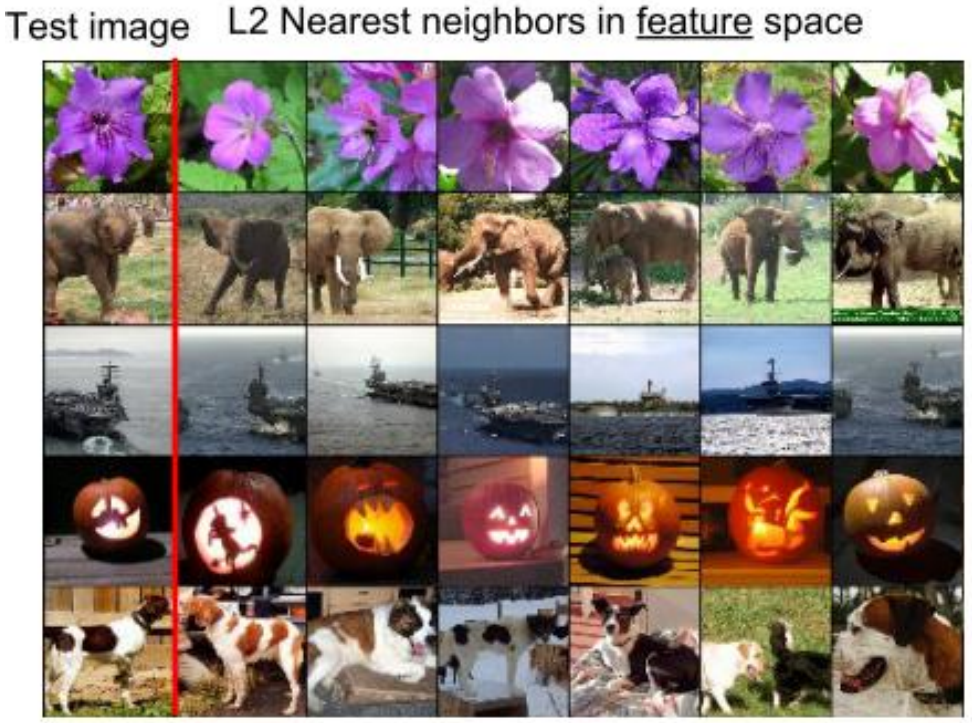




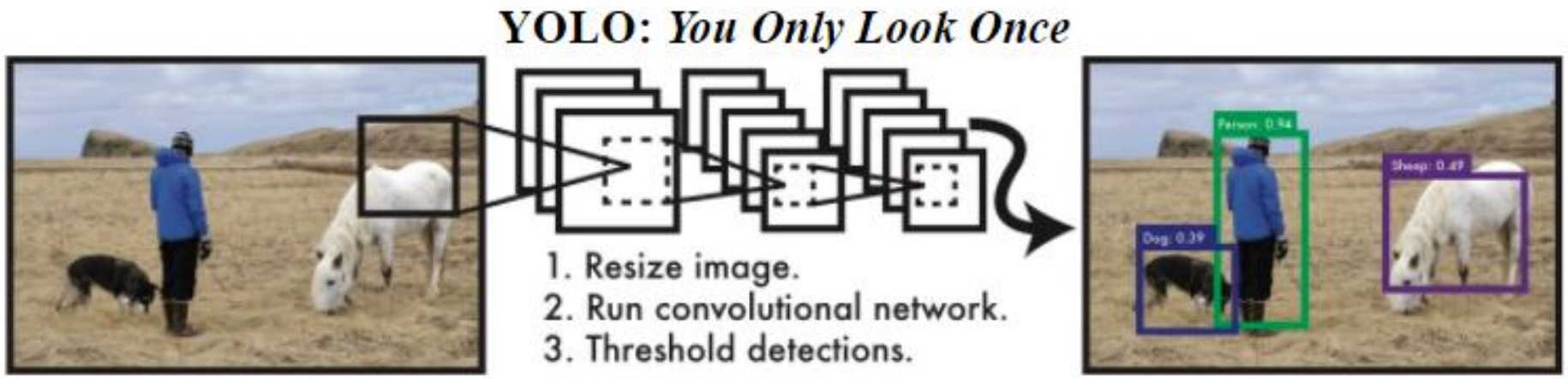
Inside the neural networks

- **Final layer**
- **Nearest neighbours**



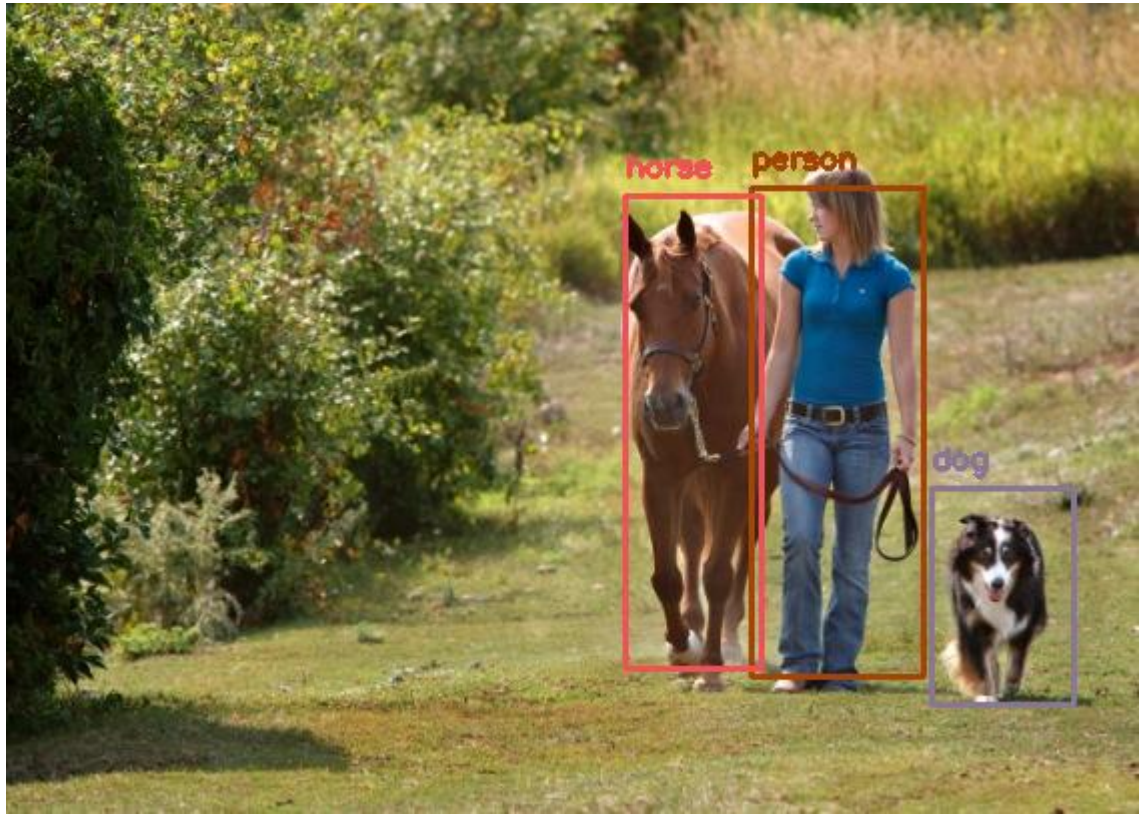


Object Detection: YOLO



YOLO: You Only Look Once

Results of Object Detection



Object detection results using YOLO3

Summary

- **Image fundamentals**
- **Face recognition**
- **Image segmentation**
- **Image recognition**